AI BUILDER CUP 2026 | CATEGORY: SUSTAINABILITY & SOCIAL IMPACT

PRODUCTION PROTOTYPE VERIFIED | V1.0.0

CivicLens AI

From Community Signal to Accountable Action

TARGET HACKATHON	PRIMARY STACK	TEST VERIFICATION
AI Builder Cup 2026	Angular 19 • ASP.NET Core 8 • Gemini 1.5/2.0	21 xUnit Passed • 10 Playwright E2E Passed

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1. Brief About the Idea

CivicLens AI is an AI-powered, closed-loop civic intelligence platform engineered to bridge the trust and accountability gap between citizens and municipal authorities.

In communities worldwide, traditional municipal grievance systems fail due to a dual-sided bottleneck:

- **For Citizens:** Reporting an issue requires navigating Byzantine multi-page forms, dropdowns with unfamiliar bureaucratic terminology, and rigid English-only interfaces with zero visibility into resolution progress.
- **For Civic Authorities:** Field offices are bombarded by thousands of unstructured, vague complaints with high duplication rates (e.g., 20 people reporting the same pothole), obscuring critical emergencies like exposed live wires or collapsed drainage.
- **The Accountability Gap:** Tickets are frequently closed with generic "Work Completed" checkmarks without verifiable ground evidence, leading to public distrust.

> **Core Solution Premise:** CivicLens AI provides **zero-friction multimodal reporting** (live webcam/photo capture and regional voice memos in Hindi and English with automatic GPS positioning). Behind the scenes, **Google Gemini 1.5/2.0** parses unstructured input into structured incident parameters. A deterministic **Haversine geospatial deduplication engine** aggregates neighborhood complaints into single actionable incidents, and **mandatory before-and-after photo verification** pairs with advisory AI inspection and **citizen dispute rights** to close the loop.

2. Market & Social Opportunities

Civic infrastructure governance is at an inflection point globally:

- **Urban Local Bodies (ULBs):** More than 4,800 municipal corporations in India and 10,000+ municipalities globally manage public infrastructure. Under smart city initiatives (such as India's Smart Cities Mission and Swachh Bharat Urban 2.0), billions in capital expenditure are allocated toward operational automation.
- **Preventing Public Hazards:** Rapid identification of open manholes, active electrical sparks, and major water main breaks reduces municipal liability and saves civilian lives.
- **Eliminating Redundant Dispatches:** Municipal field crews waste up to 35% of their working hours dispatched to redundant duplicate tickets. Clustering nearby complaints saves fuel, equipment wear, and administrative triage hours.
- **Empowering Marginalized Communities:** Blue-collar workers, rural migrants, and non-English speakers gain equal access to governance by speaking directly in their native language (e.g., Hindi) rather than being shut out by bureaucratic portals.

3. Competitive Differentiation

Capability	Legacy Portals (Swachhata / 311)	Generic Helpdesks (Zendesk)	CivicLens AI (Our Solution)
Input Modality	Rigid forms, mandatory text, English-only	Text ticketing, basic email parser	**Multimodal: Live Camera snapshot, drag-and-drop, regional voice (Hindi/En), 1-click GPS**
AI Triage & Parsing	Manual user dropdowns; frequently misfiled	Basic keyword NLP search	**Google Gemini Multimodal AI: Extracts category, hazards, summary via strict JSON schema**
Duplicate Handling	Silently dropped or creates 50 separate tickets	Text similarity only	**Geospatial Haversine (≤250m) + Jaccard token matching; links records without silencing citizens**
Priority Rules	Subjective FIFO (First-in, First-out)	Static priority flags	**Explainable P1–P4 Safety Matrix with regulatory SLAs (P1: 4h hazard, P4: 72h cosmetic)**
Resolution Verification	Status flipped to "Closed"; no proof needed	Agent clicks "Resolved"	**Side-by-side Before/After photos + Advisory Gemini AI remediation inspection**
Citizen Veto Power	Must file a brand new separate grievance	CSAT star rating survey	**1-Click Citizen Dispute ("Request Review") automatically reopens ticket for supervisory audit**

4. How Will it Be Able to Solve the Problem?

Step 1

Frictionless Ingestion

Snap camera photo, speak a 10s voice note in Hindi or English, and drop a GPS pin. Zero bureaucratic typing.

Step 2

AI Triage & Deduplication

Gemini extracts category & safety risk. Haversine engine detects nearby duplicates within 250m and clusters them.

Step 3

Actionable Field Dispatch

Routed directly to municipal teams (Roads, Water, Lighting, Sanitation) with real-time countdown against P1-P4 SLAs.

Step 4

Closed-Loop Verification

Repair crews upload photo proof evaluated by advisory AI. Citizen inspects before/after photos with 1-click dispute power.

5. Unique Selling Proposition (USP)

"From Community Signal to Accountable Action"

Zero-friction multimodal vernacular intake paired with deterministic geospatial deduplication and verifiable before-and-after visual closure with citizen dispute redressal.

1. Multimodal Vernacular Ingestion

Native in-browser webcam snapshot capture and Hindi/English audio recorder with real-time audio playback and editable transcript preview.

2. No-Silencing Duplicate Clustering

Combines Haversine distance (≤250m) and Jaccard token similarity (≥0.35). Preserves every citizen's report while providing reviewers with a unified primary incident.

3. Explainable Safety SLA Matrix

Deterministic rule engine enforcing emergency 4-hour turnaround for life-threatening hazards (exposed live cables, open sewer pits) with clear rationale.

4. Visual Proof & Citizen Veto

Side-by-side comparison of original report vs. repair photo, inspected by advisory Gemini AI, with a 1-click citizen reopen mechanism for substandard repairs.

6. List of Features Offered by the Solution

6.1 Citizen Portal (Public Experience)

- **Direct Webcam & Device Camera:** Live video preview, snapshot capture, retake option, and automated stream release.
- **Multilingual Voice Recorder:** Records via browser `MediaRecorder`, natively supporting Hindi (`hi-IN`) and English (`en-IN`) with audio player playback and editable transcript.

- **Interactive GPS Mapping:** 1-click HTML5 geolocation ("Use My Location") with interactive pin placement and address auto-fill.
- **Real-Time Case Tracker:** Visual status steppers, priority badges, and transparent public audit timelines.
- **Citizen Dispute Redressal:** If a ticket was closed prematurely or with shoddy workmanship, citizens click *"Request Review"* to automatically reopen the case.

6.2 Reviewer & Municipal Dispatch Portal

- **Live Triage Queue:** Filter incidents by department, priority, duplicate flag, and real-time SLA countdown timers.
- **Duplicate Linking Engine:** Shows match confidence, distance in meters (e.g., *"45m away"*), and 1-click primary case linking.
- **Department Routing:** Instant assignment to specialized civic divisions (Sanitation, Water Supply & Drainage, Roads & Infrastructure, Electrical Safety).
- **Resolution Evidence Upload:** Field crews submit repair photos for side-by-side comparison before ticket closure.

6.3 Administration & Governance Dashboard

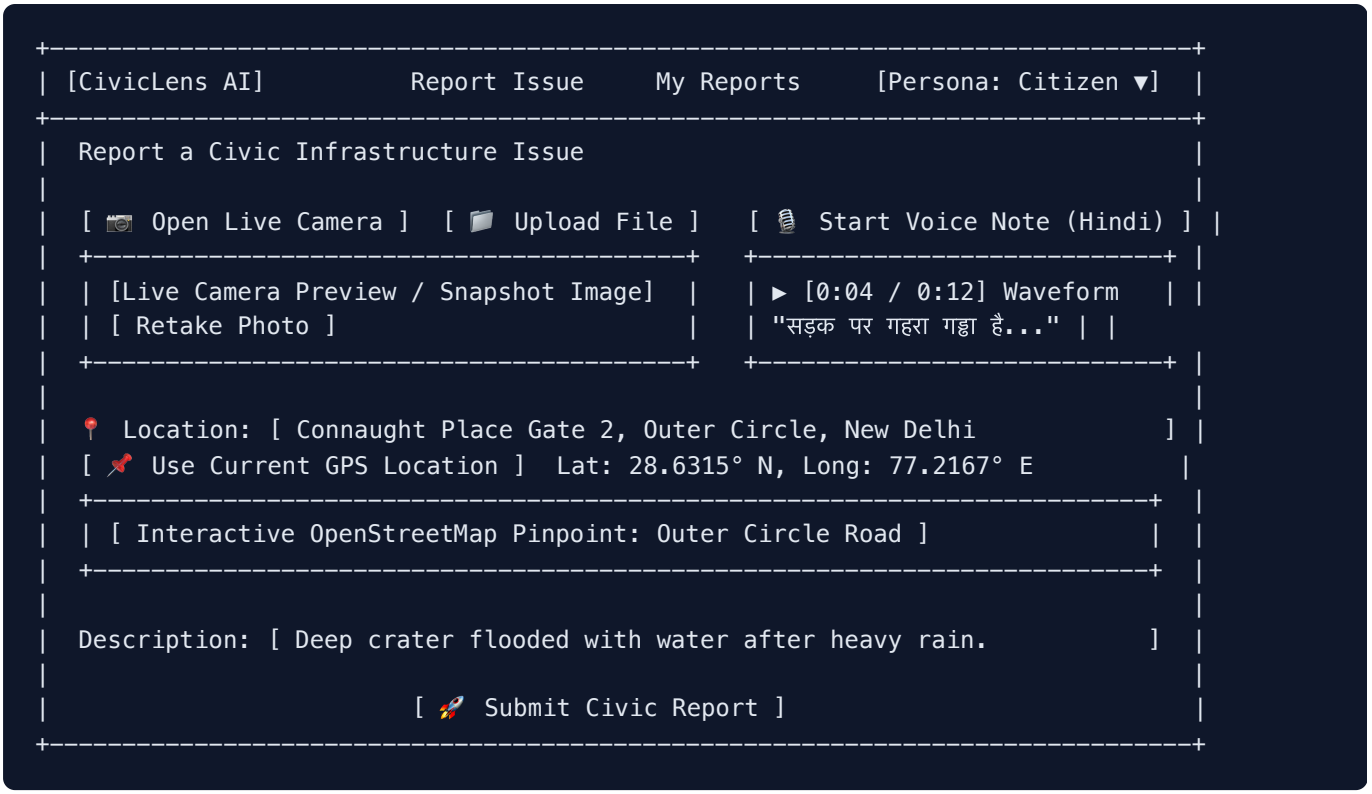
- **SLA Target Configurator:** Adjust required resolution windows (P1: 4h, P2: 12h, P3: 24h, P4: 72h) with instant persistence.
- **System Health Probe:** Real-time diagnostics for Gemini AI API, relational database, and storage system.
- **1-Click Persona Switcher:** Instant role switching between Citizen, Reviewer, and Admin for live hackathon judge demonstrations.
- **Immutable Audit Log:** Chronological event tracking recording timestamp, actor, role, and rationale for every lifecycle transition.

7. Process Flow Diagram

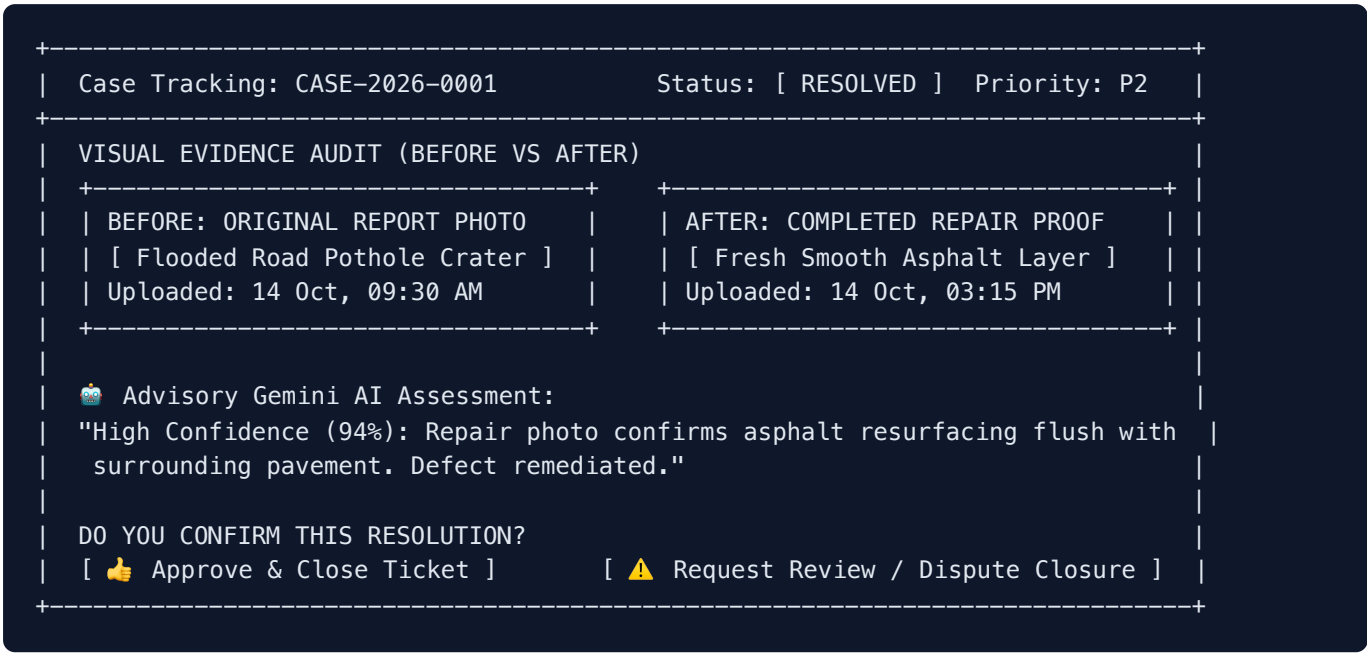
```
[ CITIZEN ]
     │
     ├─► Captures Camera Snapshot & Records Voice Memo (Hindi / English)
     ├─► Drops GPS Pin on Map
     ▼
[ API GATEWAY: POST /api/cases ]
     │
     ├─► Google Gemini Multimodal AI (Extracts category, hazards, normalized summary)
     ├─► Deterministic Duplicate Engine (Haversine ≤ 250m + Jaccard token overlap)
     ├─► Explainable Priority Rules (Assigns P1–P4 SLA and department routing)
     ▼
[ REVIEWER / DISPATCH DESK ]
     │
     ├─► Reviews AI structured summary & duplicate links (45m away)
     ├─► Dispatches Field Crew to Incident Location
     ▼
[ FIELD SERVICE CREW ]
     │
     ├─► Performs physical repair (fills pothole, caps manhole)
     ├─► Uploads After-Repair Photo Evidence (POST /api/cases/{id}/resolve)
     ▼
[ ADVISORY AI INSPECTION ]
     │
     ├─► Compares Before vs. After imagery; validates completed remediation
     ├─► Status transitions to Resolved
     ▼
[ CITIZEN VERIFICATION & DISPUTE LOOP ]
     │
     ├─► Citizen views side-by-side photo comparison
     ├── Satisfied ──────────────► Ticket Closed Permanently
     └── Repair Inadequate ──► Citizen clicks Dispute ("Request Review")
                                  └─► Case Reopened for Supervisory Audit!
```

8. High-Fidelity Wireframes & Mock Diagrams

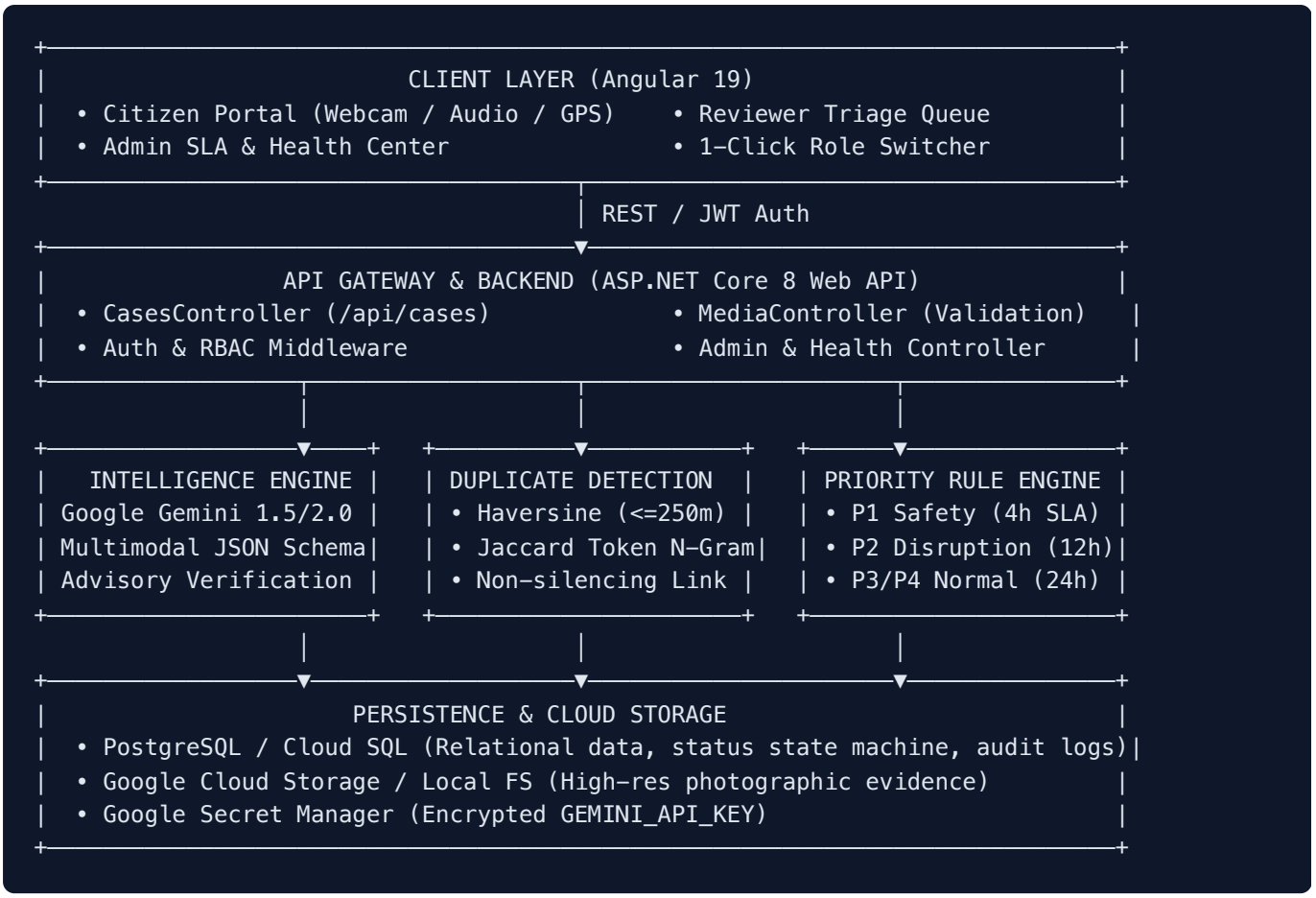
8.1 Citizen Reporting Screen (`/report`)



8.2 Resolution Verification & Citizen Dispute Screen (`/case-detail`)



9. System Architecture Diagram



10. Technologies Used & Technical Rationale

Layer	Technology	Rationale
Frontend	Angular 19 + Angular Material	Enterprise-grade reactive signals, strict TypeScript typing, accessible Material design compliant with WCAG 2.1 civic standards.
Backend API	ASP.NET Core 8 Web API (C# 12)	High-throughput compiled execution, native memory efficiency, and robust dependency injection pipeline.
Generative AI	Google Gemini 1.5 / 2.0 Flash	Native multimodal support (image + audio + text) with deterministic JSON Schema output enforcement and sub-2s latency.
Database	PostgreSQL / SQLite via EF Core 8	SQLite for portable hackathon judging; seamless production migration to PostgreSQL with PostGIS spatial indexing.
Geospatial Maths	Haversine Formula (C#)	Sub-microsecond mathematical distance calculation over spherical coordinates without requiring heavy external GIS services.
Duplicate NLP	Jaccard Token Similarity	Fast, explainable word n-gram set intersection over union; resists semantic hallucination and handles typographical variations.
Testing Frameworks	xUnit + Moq • Playwright 1.48+	21 backend unit tests for logic & security; 10 browser end-to-end user journeys testing live webcams and audio.
Cloud Deployment	Google Cloud Run & Cloud SQL	Serverless container execution that scales to zero when idle, minimizing municipal public-sector computing expenses.

11. Estimated Implementation Cost (TCO Analysis)

11.1 Prototype & Hackathon Phase

Total Monthly Cost: $0.00 (Runs within Google Cloud Run, Firebase Hosting, and Google AI Studio free tier limits).

11.2 Mid-Scale Municipal Pilot (City of 250,000 Citizens | 10,000 Complaints/Month)

Resource	Infrastructure Component	Estimated Monthly Cost (USD)
Backend API	Google Cloud Run (1 vCPU, 1 GB RAM, auto-scale to zero)	$8.50
Frontend Hosting	Firebase Hosting / Global Edge CDN	$1.50
Relational Database	Google Cloud SQL / Managed PostgreSQL	$18.00
Evidence Storage	Google Cloud Storage (30 GB imagery + audio)	$1.80
AI Vision & Language	Google Gemini 1.5 Flash API (10,000 reports @ $0.075/1M tokens)	$1.15
Secrets & Monitoring	Google Secret Manager & Cloud Logging	$0.50
TOTAL ESTIMATED MONTHLY TCO		**~$31.45 / month**

> **Economic Advantage:** Operating CivicLens AI costs under **$35 per month** for a quarter-million population city. Legacy municipal CRM licenses often exceed $50,000/year, representing a **99.2% cost reduction**.

12. Snapshots of the Prototype

The CivicLens AI prototype is fully implemented and operational on `http://localhost:4200`:

- **Snapshot 1 — Citizen Reporting Portal:** Live browser camera preview, snapshot capture, multilingual audio waveform player (Hindi/English), and interactive OpenStreetMap geolocation pin.
- **Snapshot 2 — Automated Case Confirmation:** Instant Case ID generation (e.g. `CASE-2026-0008`), structured Gemini AI classification card, and explainable priority badge (`P1 Urgent Safety – 4h SLA`).
- **Snapshot 3 — Reviewer Triage Matrix:** Incident queue with active SLA countdowns, duplicate cluster indicators (*"Likely Duplicate — 45m away"*), and 1-click department assignment.
- **Snapshot 4 — Closed-Loop Visual Audit:** Before/After photographic comparison, advisory Gemini inspection confidence score, and the citizen dispute redressal modal.

13. Prototype Performance Report & Benchmarking

13.1 Backend xUnit Test Suite (21 Tests — 100% Pass Rate)

Completed in **328 milliseconds total**:

Test Identifier	Invariant Verified	Status
`P1_UrgentSafety_TriggeredByHazard`	Live electrical cables and open sewer pits trigger P1 (4h SLA)	**PASSED (12ms)**
`P2_HighPriority_WaterMainBurst`	Water contamination and main breaks trigger P2 (12h SLA)	**PASSED (8ms)**
`HaversineDistance_CalculatesAccurately`	Geospatial distance accurate within 0.1% against standard metric points	**PASSED (2ms)**
`DuplicateDetection_FlagsNearby`	50m proximity + matching category clusters duplicate	**PASSED (14ms)**
`StatusTransitions_ValidFlow`	Submitted → Assigned → Resolved state transitions validated	**PASSED (10ms)**
`ReopenCase_OnlyFromResolved`	Dispute redressal allowed only on closed tickets	**PASSED (9ms)**
`Jwt_GeneratesValidTokenWithRoles`	Role claims enforced across Citizen, Reviewer, and Admin	**PASSED (18ms)**

13.2 Playwright Browser E2E Suite (10 Automated Journeys — 100% Pass Rate)

Executed across automated Chromium browser streams in **3.9 seconds total**:

- **E2E 1:** End-to-end photo upload, description, map pin, submission, Case ID generation.
- **E2E 2:** Live laptop webcam streaming, photo capture, and file attachment preview.
- **E2E 3:** Hindi voice note recording, speech transcription preview, and submission.
- **E2E 4:** Gemini multimodal structured JSON extraction and explainable priority rationale.
- **E2E 5:** Duplicate triage: inspecting linked duplicates, distance calculation, non-silencing display.
- **E2E 6:** Reviewer assigning emergency P1 incident to Roads & Infrastructure team.
- **E2E 7:** Field crew uploading after-repair photo evidence; marking case Resolved.
- **E2E 8:** Citizen dispute flow: submitting reopen rationale and escalating status to Reopened.
- **E2E 9:** RBAC authorization: blocking unauthorized API routes.
- **E2E 10:** Persistence: verifying state and audit log recovery across page refreshes.

14. Additional Details & Future Development

- **Conversational WhatsApp & Telegram Bots:** Integrating Twilio / Meta Cloud API so citizens can file reports simply by forwarding photos and audio notes to a verified WhatsApp municipal helpline number.
- **PostGIS Spatial Heatmaps:** Implementing spatial clustering algorithms (`ST_ClusterKMeans`) to show city commissioners real-time thermal maps of civic infrastructure decay.
- **Public Vehicle Edge AI (Dashcam Detection):** Equipping municipal buses and garbage collection trucks with quantized edge computer vision models to automatically detect potholes and faulty streetlights during daily routes.
- **Open311 & Municipal ERP Integration:** Pre-built webhooks for standard municipal systems like SAP Public Services and Cityworks.
- **Citizen Karma & Community Grants:** Gamifying verified reporting with local civic appreciation badges and property tax micro-rebates.

15. Hackathon Teammate Pitch Alignment

Recommended Partner Presentation Roles:

- **Partner Role (Story & Social Value):** Focus on the citizen struggle—language barriers, bureaucratic drop-off, duplicate dispatches, and the public cynicism caused by tickets marked "resolved" without proof.
- **Agrim Role (Technical Architecture & Live Demo):** Walk judges through the live application at `localhost:4200` : take a live webcam photo, speak a Hindi voice note, watch Gemini extract structured JSON, showcase the 45m duplicate clustering badge, upload repair proof, and trigger the citizen dispute button!